Mesiri

Field Guide · 03 of 09 · Internal team edition

Materials

How a construction company receives, stores, issues and accounts for material; where the money leaks; what good control looks like; and what a complete system must do.

Draft, 7 Nov 2026. Written from industry practice, construction ERPs and customer meetings. Practices, documents and numbers vary between companies and must be checked by someone with site experience before we use them with customers.

How to read this guide. Parts 1 to 7 are about the business, not about any software. Part 8 compares Mesiri with the checklist in part 4: that is the list of what to build next.

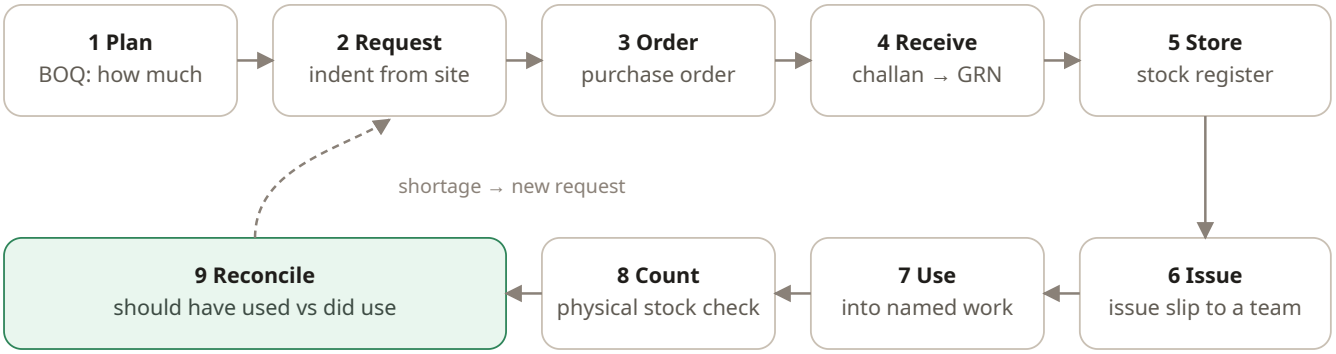
In this guide

1. How materials work on a construction project
2. The problems: where money leaks
3. What good control looks like
4. What a complete system needs
5. Best UX for materials software
6. Where WhatsApp and voice fit
7. How it connects to other modules
8. Mesiri today: the gap list
9. Key terms · Questions · Check yourself

1 · HOW MATERIALS WORK ON A CONSTRUCTION PROJECT

From "we need it" to "it is in the wall"

Material is usually the largest cost on a building project, often around half of it. It is bought in many small lots from many suppliers, delivered to several sites, stored in the open or in a small shed, and used a little at a time by different teams. Every step has a person responsible and, in a well-run company, a document.



Side paths: returns to supplier · transfers between sites ·

The steps, the document, and who is responsible

STEP	WHAT HAPPENS	DOCUMENT	WHO
1 Plan	The estimate (BOQ) gives quantities of work; standard material per unit of work turns them into material needed: e.g. 1 m^3 of M20 concrete needs roughly 8 bags of cement plus sand and metal.	BOQ, material schedule	QS, planning engineer
2 Request	The site asks for what the next days' work needs, with a needed-by date.	Indent / material requisition	Site engineer; approved by PM
3 Order	Purchase picks a supplier and agrees quantity, rate, GST, delivery date and terms.	Purchase order (PO)	Purchase / PM
4 Receive	The lorry arrives with a delivery challan. At the gate the material is counted, measured or weighed, checked for quality, and short or damaged quantity is rejected. What is accepted is written up.	Delivery challan, gate entry, GRN (goods receipt note), weighbridge slip	Storekeeper / security; engineer for quality
5 Store	Accepted material goes into stock: cement stacked off the ground, steel by diameter, sand and metal in marked heaps.	Stock register / bin card	Storekeeper
6 Issue	Material leaves the store to a team or subcontractor for a named piece of work, signed for.	Material issue slip (MIN)	Storekeeper; receiver signs
7 Use	It goes into the work. Leftovers come back to the store.	Daily report, return slip	Foreman, engineer
8 Count	Stock is physically counted every week or month and compared with the register; differences are adjusted with a reason.	Stock verification sheet	Storekeeper, checked by PM or accounts

STEP	WHAT HAPPENS	DOCUMENT	WHO
9 Reconcile	For the work actually done, how much material should have been used? Compare with what was used. Excess beyond an agreed wastage allowance is investigated.	Material reconciliation statement	QS

Side documents: **return to supplier** (debit note), **site-to-site transfer** (transfer note or internal challan), **wastage / scrap register** (broken blocks, hardened cement, steel offcuts sold as scrap).

Material behaves differently by type

TYPE	EXAMPLES	HOW IT IS RECEIVED	SPECIAL CARE
Countable	Cement bags, blocks, tiles, pipes	Counted at the gate	Cement loses strength with age: use oldest first (FIFO), check the manufacture date
Bulk, measured	Sand, M-sand, metal (aggregate)	Lorry body measured (length × width × height), in cft or brass	Cannot be counted once dumped: only receipts and use can tell you what should be in the heap
Weighed	Steel (TMT bars), binding wire	Weighbridge slip; bars counted by diameter	Test certificate from the mill; offcuts and scrap
Small, many types	Electrical and plumbing fittings, hardware	Counted against the order, item by item	Many sizes; leftovers disappear; drawn up from the MEP drawings
Free issue	Cement or steel supplied by the client to the contractor	As above, but owned by the client	Contracts often require reconciliation; excess use beyond the allowance can be recovered from the contractor

Money leaves the company as material, and nobody sees where it goes

> *"Near project closeout, you suddenly discover a shortage of 4 loads of metal. Reflected on paper, but missing on the ground. Accounts are tied to that: if money left, the material should be there."*
>
> GM of a Kerala builder, sales meeting, 6 Nov 2026

Leaks at each step

STEP	WHAT GOES WRONG	WHAT IT COSTS, AND WHO FEELS IT
Request	Site asks late, or two people ask for the same thing. The electrician asks for 10 pipes and uses 6.	Urgent buying at worse rates; duplicate stock; leftovers that vanish.
Order	No comparison of rates; nobody notices cement is ₹14 a bag dearer than last month.	A few percent on every purchase, every month. Owner and purchase.
Receive	Short delivery accepted by the challan, not by counting. A lorry of sand "looks full". Steel not weighed.	The company pays for material it never received. Accounts pays the full bill; nobody can prove otherwise later.
Store	Cement stored on the ground gets damp; old bags used last; steel rusts; no lock on the store.	Spoiled material and theft. Storekeeper and PM.
Issue	Material taken without an issue slip; nobody knows which team took it or for which work.	Use cannot be checked against work; subcontractors over-draw.
Use	More used than the work needs: thick plaster, poor mixing, breakage.	The job's margin shrinks quietly. QS finds out at the end.
Transfer	20 bags taken to another site in the company's lorry, never written down.	One site looks short, another has extra nobody knows about.
Count	No regular stock check; the register drifts away from the ground.	Losses found months later, when nobody remembers why.
Reconcile	Nobody compares material used with work done.	The "4 loads missing at closeout" problem. Too late to recover.

Rough size of a leak. If a load of 20 mm metal is about 200 cft (lorry sizes vary) at ₹46 a cft, the GM's 4 missing loads are about **₹36,800** on one material, on one site. A 3% leak on ₹1 crore of material is ₹3 lakh per project. These are illustrations, not measured figures.

The problems people have with materials software

Contractors have tried registers, Excel and apps. Most stop using them, for the same reasons:

- **Too much typing on site.** A storekeeper with dusty hands will not fill a 10-field form for every lorry.
- **Units that don't add up.** Sand comes in loads, is used in cft or bags; steel in tonnes, used in bars.
- **One item, five names.** "M sand", "msand", "manal" become separate items and totals split.
- **No opening stock**, so the numbers are wrong from day one and nobody trusts them after.
- **Only deliveries are entered, never use**, so stock only ever goes up.
- **Silent edits** change last week's totals and nobody knows why.
- **Price and bill are not known at the gate**, so the form can't be completed and is skipped.
- **A register, not an answer.** Someone must read pages of entries to find the one problem.

3 · WHAT GOOD CONTROL LOOKS LIKE

The habits of contractors who don't lose material

Count at the gate, not from the challan.
Count bags, measure the lorry body for sand and metal, weigh steel. Accept what is counted; note what is rejected.

Separate the hands.
The person who orders is not the one who receives, and neither is the one who pays. The bill is paid only when order, delivery and bill agree (3-way match).

Issue against work.
Material leaves the store for a named piece of work and a named team, signed for.

Oldest first.
Cement and other perishables are used in the order they arrived.

Count regularly.
Weekly for high-value items, monthly for the rest; surprise checks now and then.

Reconcile against the estimate.
For the work done, what should have been used? Agree a wastage allowance per material and investigate anything above it.

Control by value (ABC).
Tight control of the few expensive items (cement, steel, tiles); light control of cheap ones (nails, binding wire).

Every correction leaves a trace.
Who changed what, when and why.

The numbers good companies watch

MEASURE	WHAT IT MEANS	EXAMPLE
Days of cover	How many days current stock lasts at the recent rate of use.	Cement: 217 bags, about 10 days.
Wastage %	Used beyond what the work needed, against the allowance agreed for that material.	Cement 4% against a 2% allowance: investigate. (Allowances vary by company and contract.)

MEASURE	WHAT IT MEANS	EXAMPLE
Material variance (₹)	Value of material used above or below the estimate for the work done.	Plastering this month: ₹18,000 over.
Stock accuracy	How close the physical count is to the register.	Count 205 bags, register 217: 12 bags to explain.
Rate variance	Price paid against the budget or last rate.	Cement ₹414 against ₹400 budget: +3.5%.
Stock-outs	Days work stopped or slowed for lack of material.	2 days in October.
Unbilled deliveries	Material received with no supplier bill yet.	6 deliveries, ₹1.2 lakh.

4 · WHAT A COMPLETE SYSTEM NEEDS

The requirements checklist

Drawn from how the steps in part 1 are run and from what construction ERPs and site apps offer. **Must** = a contractor will not adopt without it. **Should** = expected by a serious buyer. **Nice** = differentiator or for larger companies.

#	REQUIREMENT	PRIORITY
A · Item list (masters)		
A1	Item list per company with category, sub-category and one stock unit per item	Must
A2	Other names for the same item (aliases), and search by any of them	Must
A3	Unit conversions: standard ones built in (tonne/kg, brass/cft, cum/cft, cement bag/kg), and company sizes ("1 load = 60 cft")	Must
A4	Grade, brand and size as part of the item (OPC 53, Fe500D, 12 mm)	Should
A5	Minimum / reorder level per item and site	Should
A6	Import items, categories and opening stock from Excel	Must
A7	Value class (ABC) to decide how tightly an item is controlled	Nice
B · Receiving		
B1	Record a delivery: site, item, quantity, date, who received it	Must
B2	Supplier and rate, optional at the gate, completed from the bill later	Must
B3	Delivery against a purchase order, including part deliveries and what is still pending	Must
B4	Photo of the challan and of the loaded lorry	Must
B5	Accepted and rejected quantity, with the reason	Should
B6	GRN number; vehicle number; weighbridge slip or measured lorry volume	Should
B7	Quality at receipt: test certificate, manufacture date (cement)	Nice
C · Stock		
C1	Live stock per item and site, and across all sites	Must
C2	Stock for a period: opening, received, used, adjusted, closing	Must
C3	Days of cover and a running-low alert	Must
C4	Negative stock shown and flagged, never hidden	Must
C5	Physical stock count with an adjustment and reason for each difference	Must
C6	Stock value (at last rate or average rate)	Should
C7	Batches and oldest-first for perishables (cement)	Nice
D · Issue and use		
D1	Record use with the work it went into	Must
D2	Use linked to a BOQ line, so it can be compared with the work done	Must

#	REQUIREMENT	PRIORITY
D3	Issue to a named team or subcontractor, signed for	Should
D4	Returns from site back to the store	Should
D5	Free-issue material (client's) tracked separately, with recovery for excess use	Should
E · Transfers, returns, wastage		
E1	Site-to-site transfer as one record (out of one, into the other)	Must
E2	Return to supplier, with a debit note	Should
E3	Wastage and scrap register (and scrap sales)	Should
E4	Central store or yard that supplies sites	Nice
F · Requests and ordering		
F1	Request from site with a needed-by date and the work it is for	Must
F2	Approval of requests; warning when another site already has the stock	Should
F3	Duplicate request / order warning	Should
G · Reconciliation and audit		
G1	Should-have-used (BOQ × material per unit + allowance) vs used, per material and work	Must
G2	Variance report in quantity and ₹, with wastage allowances per material	Must
G3	Full history: every entry with who and when; corrections as reversals, never silent edits	Must
G4	Lock a closed period	Nice
H · Reports, alerts, access		
H1	Spend by category, supplier, site and project	Should
H2	Rate history per item and supplier	Should
H3	Weekly / monthly stock report (PDF and Excel), sent on schedule	Should
H4	Export to Excel	Must
H5	Roles: who records, who corrects and adjusts, who sees rates	Must
H6	Recording by chat or voice, without an app	Should

5 · BEST UX FOR MATERIALS SOFTWARE

What makes site people use it, and what makes them give up

Do

- Ask for the minimum at the moment it happens: item and quantity. Everything else can come later.
- Put the unit next to every number: "217 bags", never "217".
- Show days of cover, not only quantity. It is what a PM acts on.
- Make problems loud: negative stock in red, running-low on top.
- Let every total open the entries behind it.
- Group by category so fittings don't bury cement.
- Mark unknown prices as "not recorded".
- Answer the three questions separately: what we have, what happened, what we use.

Don't

- Make site staff install an app, log in or fill a long form.
- Let people type units freely ("bag", "bags", "bg").
- Guess a "close" item name; ask instead.
- Show stock without saying for which site and date.
- Allow silent edits of past entries.
- Show ₹0 when the price is simply not known.
- Fill one screen with 20 columns and codes.
- Make the office the only place information enters.

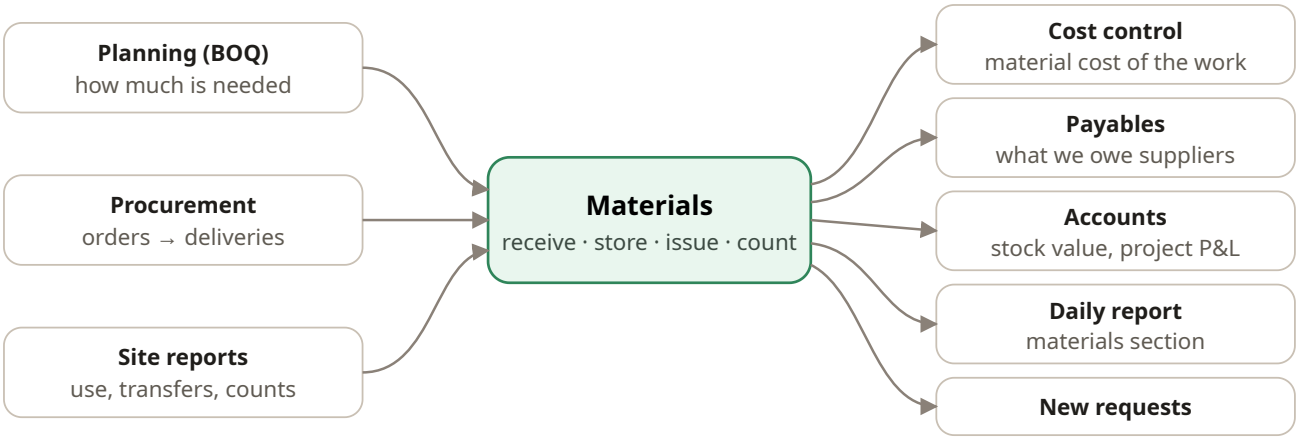
6 · WHERE WHATSAPP AND VOICE FIT

Chat for the moment it happens; a screen for checking and setup

BEST IN CHAT OR VOICE (SITE, ON THE MOVE)	BEST ON A SCREEN (OFFICE, AT A DESK)
Reporting a delivery, with a photo of the challan	Setting up the item list, categories and units
Reporting use: "45 bags into the footings"	Reconciliation: comparing use with the estimate
Transfers between sites	Physical stock count across many items
Quick questions: "how much cement is left?", "what is running low?"	Reports, trends, Excel exports
Requests: "need 200 bags by Friday"	Approving many requests or orders at once
Alerts pushed to the PM: running low, unrecorded delivery	Correcting mistakes in bulk; audit history

Rule of thumb: the person standing next to the material uses chat; the person deciding about money uses a screen. Every chat entry should be confirmed before it is saved, and the software, not the AI, should check units and item names.

7 · HOW IT CONNECTS TO OTHER MODULES



8 · MESIRI TODAY: THE GAP LIST

The checklist from part 4, against what Mesiri does now

Have works today · Partly some of it · Missing not built. Tracker IDs refer to `docs/plans/V1_TRACKER.md`.

#	REQUIREMENT	PRIORITY	MESIRI	NOTE
A1	Item list, categories, stock unit	Must	Have	Standard builder categories (PRC-01..03)
A2	Aliases and search	Must	Have	
A3	Unit conversions	Must	Have	Standard built in; company sizes asked once
A4	Grade, brand, size	Should	Partly	Only as part of the item name
A5	Reorder level	Should	Partly	Running low from rate of use; no set minimum
A6	Excel import	Must	Have	Items, categories, opening stock (PRC-05)
A7	ABC class	Nice	Missing	
B1	Record a delivery	Must	Have	WhatsApp and dashboard
B2	Supplier and rate optional	Must	Have	Vendor linked (PRC-04)
B3	Against a purchase order	Must	Missing	PRC-06, PRC-08
B4	Challan / lorry photo	Must	Partly	Supplier bills with photos exist; a photo on the delivery itself is to be checked
B5	Accepted and rejected quantity	Should	Missing	
B6	GRN no., vehicle, weighbridge	Should	Missing	Free-text notes only
B7	Quality at receipt	Nice	Missing	
C1	Live stock, all sites	Must	Have	
C2	Stock for a period	Must	Have	PRC-17
C3	Days of cover, running low	Must	Have	
C4	Negative stock flagged	Must	Have	
C5	Physical count with adjustments	Must	Partly	Adjustment entries exist; no count sheet workflow
C6	Stock value	Should	Partly	Spend shown; no valuation of what is on hand
C7	Batches, oldest first	Nice	Missing	
D1	Use with the work	Must	Have	"Used for" in words
D2	Use linked to a BOQ line	Must	Missing	BOQ exists (CC-02); link is CC-08
D3	Issue to a named team	Should	Missing	
D4	Returns from site	Should	Partly	Return entry types exist

#	REQUIREMENT	PRIORITY	MESIRI	NOTE
D5	Free-issue material	Should	Missing	
E1	Site-to-site transfer	Must	Have	
E2	Return to supplier, debit note	Should	Partly	Return recorded; no debit note
E3	Wastage and scrap	Should	Partly	Wastage entry type; no scrap sales
E4	Central store	Nice	Partly	A yard can be a "site" today
F1	Requests from site	Must	Missing	Planned page: Materials › Requests
F2	Request approval, other-site stock warning	Should	Missing	
F3	Duplicate warning	Should	Missing	PRC-07
G1	Should-have-used vs used	Must	Missing	CC-03 (norms), CC-08
G2	Variance with allowances	Must	Missing	CC-08
G3	Full history, reversals	Must	Have	
G4	Period lock	Nice	Missing	
H1	Spend by category, supplier, site	Should	Partly	By category yes; by supplier is PRC-19
H2	Rate history	Should	Partly	Last rate only (PRC-14)
H3	Scheduled weekly / monthly report	Should	Missing	PRC-19, PRC-20
H4	Export to Excel	Must	Have	CSV from Movements
H5	Roles	Must	Partly	Item list writes not role-checked in the API (known gap)
H6	Chat and voice	Should	Have	Malayalam, Manglish, English

What to build next, by this list. The missing **Must**s are: deliveries against purchase orders (B3), use linked to the BOQ (D2), requests from site (F1), and reconciliation with variance (G1, G2). Close behind: the challan photo on the delivery (B4), a stock-count workflow (C5) and role checks (H5). Most of these are already in the tracker (PRC, CC). Together they turn Materials from a register into the control the GM asked for.

9 · KEY TERMS

Indent / requisition	A request from site for material, with quantity and date needed.
Purchase order (PO)	The order to a supplier: item, quantity, rate, GST, delivery date.
Delivery challan	The supplier's delivery note that comes with the lorry.
GRN / MRN	Goods (material) receipt note: the company's record of what was accepted.
Weighbridge slip	The weight of the loaded and empty lorry; the difference is the material.
Stock register / bin card	Running record of receipts, issues and balance per item.
Material issue slip (MIN)	Record of material given from the store to a team, signed.
Opening stock	What was on hand when counting started.
Wastage allowance	The extra use above the estimate that is accepted as normal, per material.
Reconciliation	Should-have-used (for the work done) compared with actually used.
Free-issue material	Material supplied by the client to the contractor, to be accounted for.
3-way match	Order, delivery and bill agree before the bill is paid.
FIFO	First in, first out: use the oldest stock first.
ABC analysis	Controlling items by value: A expensive and tight, C cheap and light.
CFT, cum, brass	Volumes: 1 cum ≈ 35.3 cft; 1 brass = 100 cft.

QUESTIONS CUSTOMERS ASK

"Can it handle electrical and plumbing detail: fittings, bends, wire lengths?"

A good system treats each size or type as its own item under a category, so detail doesn't clutter the cement and steel.

"How do you count sand that's already dumped?"

You don't. You measure each lorry at the gate and record use; the difference is what should be in the heap, and a periodic estimate of the heap shows any gap.

"What if the engineer forgets to report use?"

Stock stays too high. Reconciliation against the work done catches it: work progressed but no material was used.

"Can we see all sites together, and move stock between them?"

Yes in a good system: one view across sites, and a transfer as one record so nothing is lost between them.

"Who can change the numbers?"

Nobody should edit history. Corrections are new entries with who and why.

CHECK YOURSELF

1. Name the nine steps of the material cycle and the document for steps 2, 3, 4 and 6.
2. Why can't you count sand once it is dumped, and how do you control it instead?
3. What is a 3-way match, and which leak does it stop?
4. Cement used was 4% above what the work needed. What do you compare that with before acting?
5. Which four missing "Must" items in part 8 would most change what the GM sees?

Answers: 1. Plan, request, order, receive, store, issue, use, count, reconcile; indent, purchase order, challan/GRN, issue slip. 2. It can't be counted in a heap; measure each lorry and record use, then compare with the heap. 3. Order, delivery and bill must agree before payment; it stops paying for material that never came or at the wrong rate. 4. The wastage allowance agreed for cement. 5. Deliveries against orders, use linked to the BOQ, requests from site, reconciliation with variance.